Diagnostic criteria for DH

The diagnosis of DH can be made if both major diagnostic criteria are fulfilled:

1. Clinical manifestation compatible with DH
2. Positive DIF microscopy

If the clinical manifestation is incompatible, no diagnosis can be made. If DIF microscopy result is repeatedly negative, but the clinical manifestation is typical for DH, the diagnosis of DH can be supported by the combination of the following minor criteria:

- Traditional histology is compatible with DH
- At least one serological test of high diagnostic value (TG2, TG3, or EMA) is positive
- Duodenal biopsy shows evidence for CD
- The result of HLA testing is compatible with DH
- Positive iodine patch test or oral iodine challenge
- Swift response to dapsone (partial recovery within 1 week of $100\, \mathrm{mg}$ daily)
- Response to a long-term gluten-free diet

HLA-DQ2/DQ8 typing is not recommended to be used routinely in the diagnosis of DH. Nevertheless, HLA-DQ2/DQ8 testing may be considered as an extension of the basic diagnostic programme for DH to effectively exclude the disease in selected clinical situations.

However, current PCR methods often investigate only the SNPs of the common alleles of DQ2.5, DQ2.2, and DQ8 (which is cheaper than traditional typing), so variant DQ2 or DQ8 alleles in certain populations may give false negative results. Therefore, the negative predictive value of such results is not as strong as earlier anticipated. Moreover, recent observations show that incomplete alleles and DQ9 also confer risk in rare patients. In line with this, DR4 or DR9 (effectively associated with DQ8 and DQ9, respectively) were described in 13/14 Japanese DH cases with granular IgA deposition in the skin.

Screening for co-morbidities

Several conditions, including autoimmune diseases such as thyroidopathies, type 1 diabetes mellitus, Addison disease, and polyendocrine syndromes, as well as lymphoproliferative diseases and hyposplenism, have been shown to have increased prevalence in patients with CD and DH. Among them, subclinical thyroid disease is the most common, and testing is relevant; thus, all patients with DH should be screened for thyroid disease (at least by measuring TSH). As lymphoproliferative disorders are also associated with DH, appropriate clinical monitoring is warranted.

Additionally, gluten-free products frequently contain higher levels of lipids, sugar, and salt to improve palatability and consistency, which may lead to excessive consumption of hypercaloric and hyperlipidic foods to compensate for dietary restrictions in patients with DH. This may negatively impact cardiometabolic risk factors such as obesity, serum lipid levels, insulin resistance, metabolic syndrome, and atherosclerosis. Furthermore, dapsone treatment may have severe side effects in patients with cardiovascular disease.

HLA haplotypes genotyping

HLA-DQ2/DQ8 typing is not recommended routinely in the diagnosis of DH.

Indication for HLA-DQ2/DQ8 testing